

# Individual Assignment 5

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## 1. Set up

### Packages

```
# general use and cleaning
library(tidyverse)
library(here)
library(janitor)
library(snakecase)

# wrapping axis labels
library(scales)

# reading in .xlsx files
library(readxl)

# visualizing missing data
library(naniar)

# arranging plots
library(patchwork)
```

## Data

## Import Graphs

## Question 1

### A: Planning

- x axis: Site
- y axis: log transformed Ostracod abundance
- geometry abundance: geom\_jitter()
- geometry median: stat\_summary()

### B: Wrangle Data

```
#new object from aq_ins_clean
ostracod <- aq_ins_clean |>
  # filter to only include ostracod
  filter(taxon == "ostracod") |>
  # log + 1 transform mean abundance per liter
  mutate(log_ave_lit = log(ave_lit + 1))

# display 10 random rows
slice_sample(ostracod, n=10)
```

```
# A tibble: 10 x 24
```

|    | date_site<br><chr> | date<br><dtm>       | site<br><chr> | taxon<br><chr> | ave_lit<br><dbl> | med_ph<br><dbl> | med_sal<br><dbl> | med_do<br><dbl> |
|----|--------------------|---------------------|---------------|----------------|------------------|-----------------|------------------|-----------------|
| 1  | 2023-06-06_NPB1    | 2023-06-06 00:00:00 | NPB1          | ostr~          | 0.533            | 7.48            | NA               | 2.38            |
| 2  | 2022-11-19_NDC     | 2022-11-19 00:00:00 | NDC           | ostr~          | 6                | NA              | NA               | NA              |
| 3  | 2021-08-25_NPB     | 2021-08-25 00:00:00 | NPB           | ostr~          | 581.             | 7.52            | 4.57             | 4.5             |
| 4  | 2022-01-21_NEC     | 2022-01-21 00:00:00 | NEC           | ostr~          | 0                | NA              | NA               | NA              |
| 5  | 2023-03-03_NDC     | 2023-03-03 00:00:00 | NDC           | ostr~          | 5.73             | 7.45            | 0.688            | 5.64            |
| 6  | 2022-02-25_NEC     | 2022-02-25 00:00:00 | NEC           | ostr~          | 0                | 8.75            | 44.1             | 9.87            |
| 7  | 2022-11-12_NEC     | 2022-11-12 00:00:00 | NEC           | ostr~          | 0                | 7.33            | 7.01             | 2.4             |
| 8  | 2023-02-28_NPB1    | 2023-02-28 00:00:00 | NPB1          | ostr~          | 0                | NA              | NA               | NA              |
| 9  | 2022-11-18_NVBR    | 2022-11-18 00:00:00 | NVBR          | ostr~          | 0                | NA              | NA               | NA              |
| 10 | 2023-09-12_NVBR    | 2023-09-12 00:00:00 | NVBR          | ostr~          | 0                | 9.14            | NA               | 20.3            |

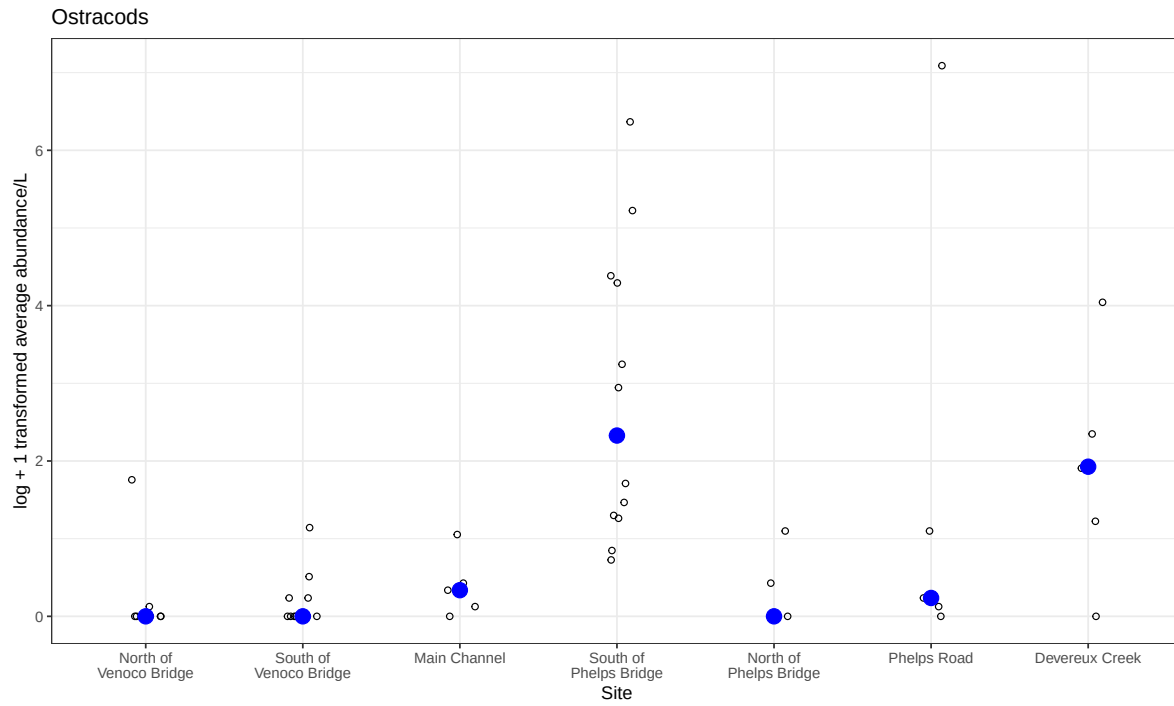
```
# i 16 more variables: taxonRank <chr>, scientificName <chr>, kingdom <chr>,
```

```
#   Phylum <chr>, Class <chr>, Subclass <chr>, Superorder <chr>, Order <chr>,
#   Infraorder <chr>, Family <chr>, Subfamily <chr>, Tribe <chr>, Genus <chr>,
#   comments <chr>, site_full <fct>, log_ave_lit <dbl>
```

## C: Code Figure

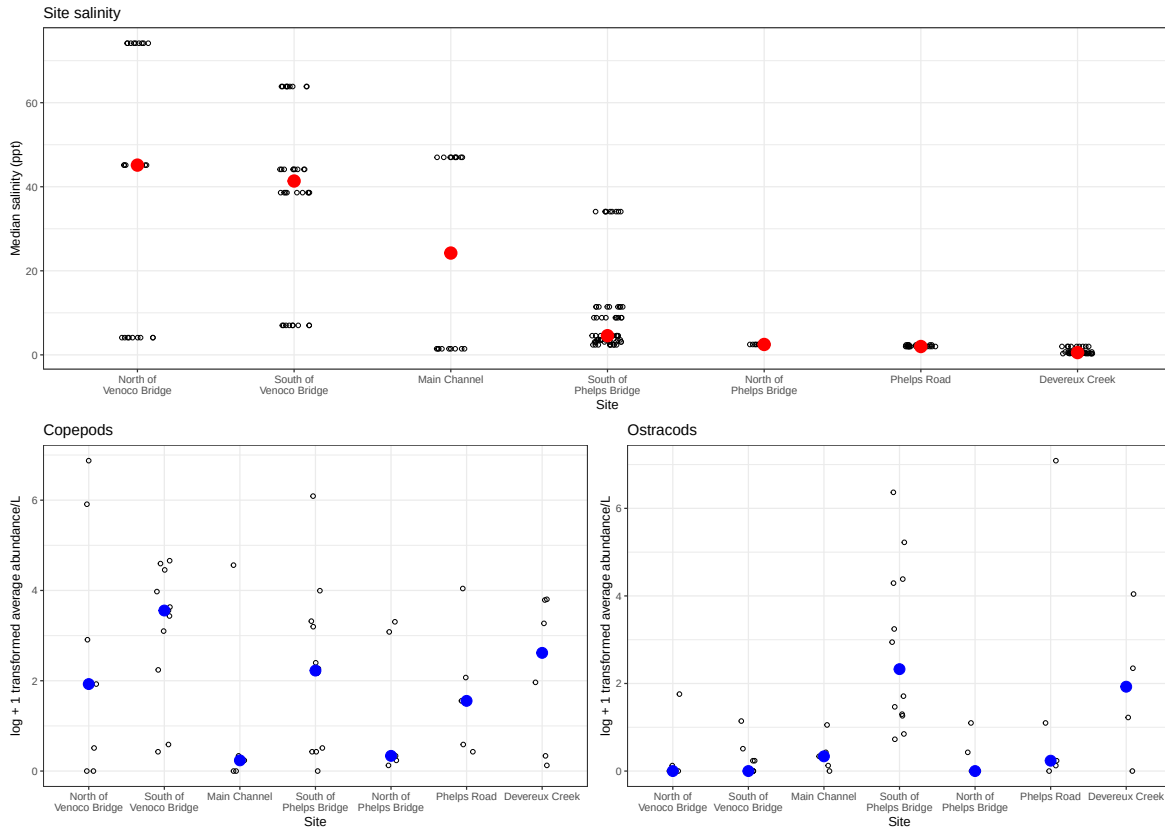
```
# base layer
ostracods_plot <- ggplot(data = ostracods,
                        # x-axis: site
                        # y-axis: mean abundance/L
                        mapping = aes(x = site_full,
                                      y = log_ave_lit)) +
# first layer: points to represent observations
geom_jitter(height = 0,
            shape = 21,
            width = 0.1) +
# second layer: points to represent medians
stat_summary(geom = "point",
            fun = "median",
            size = 4,
            color = "blue") +
# wrapping x-axis text
scale_x_discrete(labels = label_wrap(15)) +
# relabelling x- and y-axis, adding title
labs(x = "Site",
     y = "log + 1 transformed average abundance/L",
     title = "Ostracods") +
# different theme
theme_bw()

# displaying object
ostracods_plot
```



#### D: Create multipanel figure

```
#combining plots  
salinity/(copepods_plot+ostracods_plot)
```



## E: Analysis

- The differences in sites between copepods and ostracod abundance are mainly at the South of Phelps Bridge as ostracods are the most abundant here while copepods are more abundant South of Venoco Bridge.
- Ostracod abundance may not be related to site-level differences as there are numerous occasions of median salinity being relatively low at places like Phelps Bridge or Phelps road, yet South of Phelps Bridge sites have high abundance while the North side doesn't. Additionally, low abundance sites like North of Phelps have the same abundance as high salinity sites like Venoco Bridge.

## Question 2

### A: Planning

- x axis: salinity

- y axis: log transformed abundance
- color: taxon
- geometry: geom\_point()
- facets: facet\_Wrap(~taxon, scales="free")

## B: Wrangle Data

```
# creating new clean object from aquatic inverts
ab_salinity <- aq_ins_clean |>
  # log transform mean abundance per liter
  mutate(log_ave_lit = log(ave_lit+1),
    # convert taxon to title
    taxon = to_any_case(taxon, case = "title"),
    # order taxon levels by abundance
    taxon = fct_reorder(taxon, ave_lit, .fun="max"))

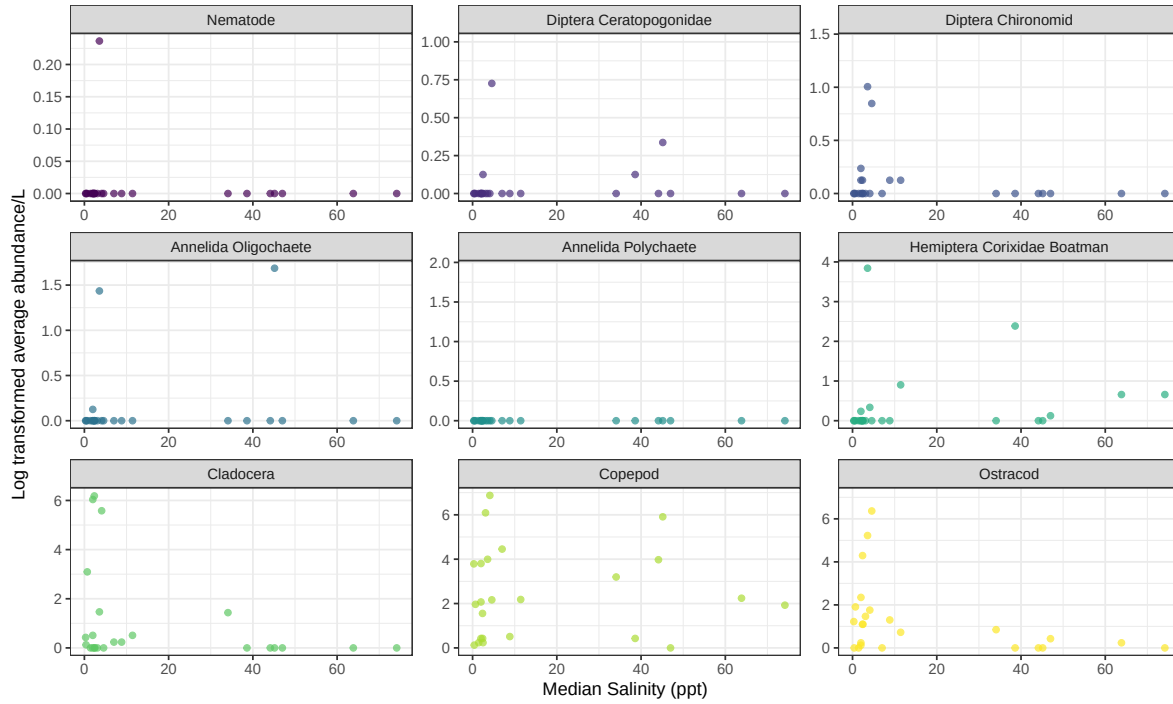
#display 10 random rows
slice_sample(ab_salinity, n = 10)
```

```
# A tibble: 10 x 24
  date_site      date      site taxon ave_lit med_ph med_sal med_do
  <chr>          <dtm>      <chr> <fct>   <dbl>   <dbl>   <dbl>   <dbl>
1 2022-01-24_NEC 2022-01-24 00:00:00 NEC Annel~    0      NA      NA      NA
2 2022-01-28_NDC 2022-01-28 00:00:00 NDC Copep~  43.9    7.65    2.00    9.55
3 2022-01-28_NDC 2022-01-28 00:00:00 NDC Dipte~    0      7.65    2.00    9.55
4 2023-05-06_NDC 2023-05-06 00:00:00 NDC Annel~    0     10.1    NA     13.1
5 2022-02-25_NEC 2022-02-25 00:00:00 NEC Hemip~    0      8.75   44.1    9.87
6 2023-11-11_NEC 2023-11-11 00:00:00 NEC Nemat~    0      NA     38.6    0.25
7 2022-08-12_NEC 2022-08-12 00:00:00 NEC Dipte~  0.133    7.37    NA      NA
8 2022-08-10_NPB 2022-08-10 00:00:00 NPB Copep~  0.667    8.28    8.84    9.81
9 2022-02-25_NEC 2022-02-25 00:00:00 NEC Clado~    0      8.75   44.1    9.87
10 2022-05-12_NPB 2022-05-12 00:00:00 NPB Ostra~   1.07    7.97   11.4    8.17
# i 16 more variables: taxonRank <chr>, scientificName <chr>, kingdom <chr>,
#   Phylum <chr>, Class <chr>, Subclass <chr>, Superorder <chr>, Order <chr>,
#   Infraorder <chr>, Family <chr>, Subfamily <chr>, Tribe <chr>, Genus <chr>,
#   comments <chr>, site_full <fct>, log_ave_lit <dbl>
```

## C: Code figure

```
# base layer
ggplot(data = ab_salinity,
      # x-axis: site
      # # y-axis: mean abundance/L
      mapping = aes(x = med_sal,
                    y = log_ave_lit,
                    color = taxon)) +

# layer 1: points to represent observations
geom_point(alpha = 0.7) +
#facet wrap for taxon
facet_wrap(~ taxon,
          scales = "free") +
# use color palette
scale_color_viridis_d() +
# labels
labs( x = "Median Salinity (ppt)",
      y = "Log transformed average abundance/L") +
# theme
theme_bw()+
# remove legend from theme
theme(legend.position = "none")
```



## D: Analysis

- The differences between taxa in relationships between abundance and salinity is that the higher the salinity, the lower the average abundance. However, this isn't the case with all species as Copepods are abundant at high and low salinities, similar to Hemiptera as both show up to 2.5 log average abundance/L in higher salinities. On the other hand, species like Annelida Oligochaete, along with the majority of the species, follow an exponential decrease as salinity increases. This includes Cladocera and Diptera Chironomid to name a few.